



**East West University**

## **Assignment-4**

**Course Code:CSE 405**

**Course Title:Computer Networks**

**Semester:Fall 2025**

**Section:05**

**Course Instructor**

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**Submitted by**

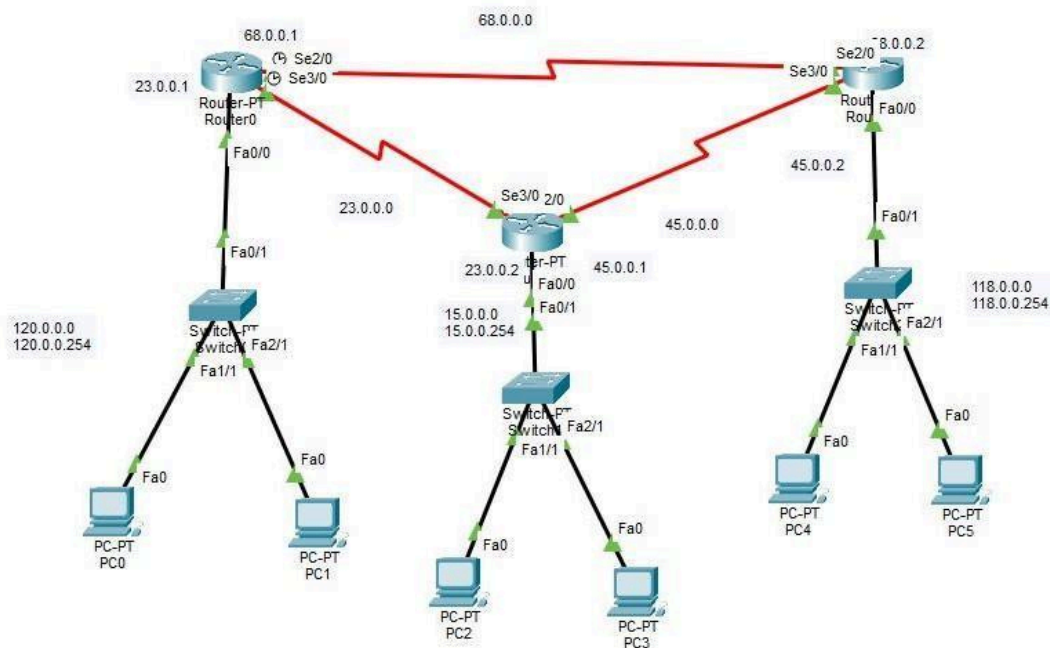
**Abdul Wadud**

**ID:2022-2-60-133**

## Introduction:

This report outlines designing a network with three routers and several LANs, as shown in the diagram. All networks receive Class A IP addresses to maintain a consistent addressing scheme. After creating the network, each router is set up with static routing so devices from different networks can communicate. The main goal of this lab is to achieve successful end-to-end connectivity. By setting up proper IP addressing, interface configuration, and static routes, PCs in different LANs can send ping messages across routers. This experiment helps to understand how routers forward packets between networks and how static routing works in a real network setup.

## Network Design:



The network consists of three routers that are connected through serial links. This setup forms a triangle-shaped backbone. Each router connects to its own LAN through a switch, and every LAN has two PCs. All networks use Class A IP addressing. Router 0 connects to the 120.0.0.0/24 LAN. Router 1 connects to the 15.0.0.0/24 LAN, and Router 2 connects to the 118.0.0.0/24 LAN. The routers are linked through separate point-to-point networks: 23.0.0.0/30 connects Router 0 and Router 1, 45.0.0.0/30 connects Router 1 and Router 2, and 68.0.0.0/30 connects Router 0 and Router 2. Static routing is set up on all routers so that each device can access every other network through the right path. With correct IP addressing, interface setup, and static routes, all PCs across different LANs can communicate successfully with each other using ping.

## Router 0: Config:

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 120.0.0.0 255.0.0.0
Bad mask /8 for address 120.0.0.0
Router(config-if)#ip address 120.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int se3/0
Router(config-if)#ip address 23.0.0.1 255.0.0.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int se2/0
Router(config-if)#ip address 68.0.0.1 255.0.0.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#exit
```

## Static routing:

```
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 15.0.0.0 255.0.0.0 23.0.0.2
Router(config)#ip route 118.0.0.0 255.0.0.0 23.0.0.2
Router(config)#ip route 45.0.0.0 255.0.0.0 23.0.0.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
|
```

## Router 1: Config:

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 15.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int se3/0
Router(config-if)#ip address 23.0.0.2 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up

Router(config)#exit
```

## Static routing:

```
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 120.0.0.0 255.0.0.0 23.0.0.1
Router(config)#ip route 118.0.0.0 255.0.0.0 45.0.0.2
Router(config)#ip route 68.0.0.0 255.0.0.0 23.0.0.1
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
|
```

## Router 2: Config:

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 118.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int se3/0
Router(config-if)#ip address 45.0.0.2 255.0.0.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int se2/0
Router(config-if)#ip address 68.0.0.2 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exi
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

%LINK-5-CHANGED: Interface Serial3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up
```

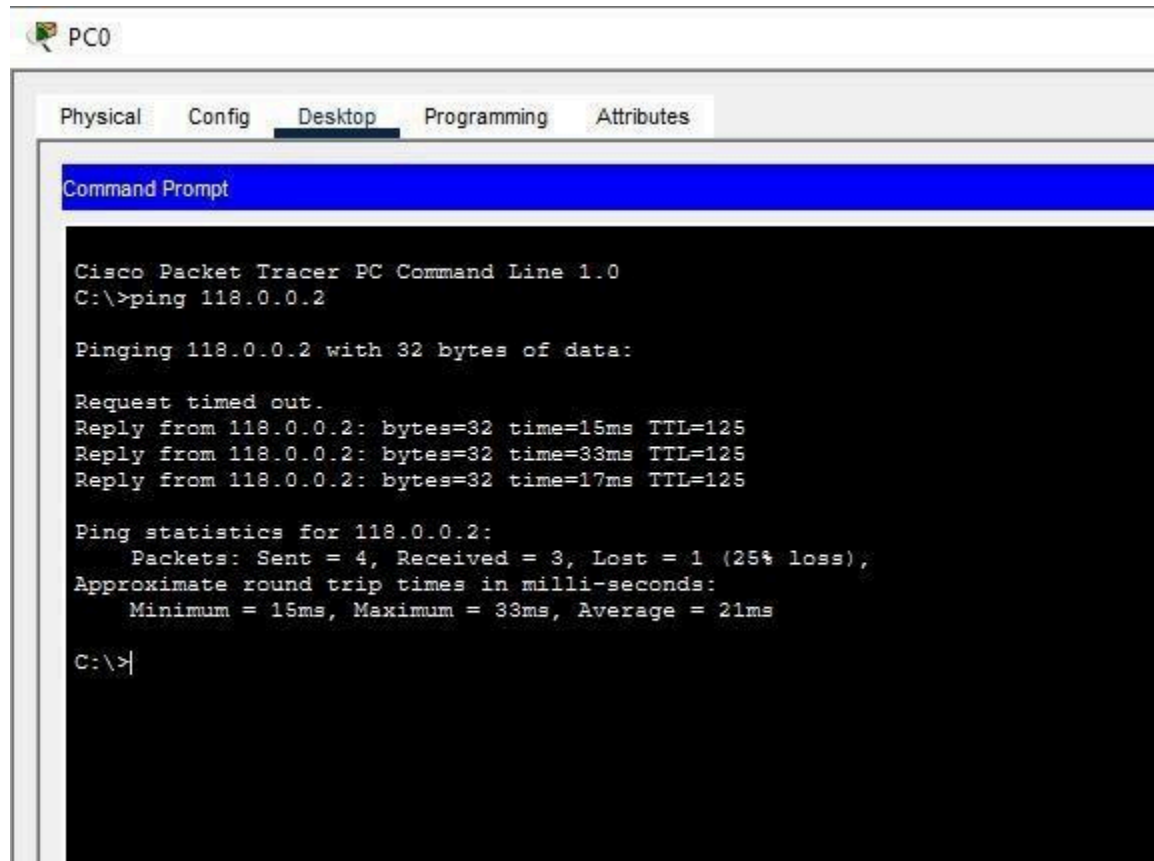
## Static routing:

```
Router#  
Router#config  
Configuring from terminal, memory, or network [terminal]?  
Enter configuration commands, one per line. End with CNTL/Z.  
Router(config)#ip route 15.0.0.0 255.0.0.0 45.0.0.1  
Router(config)#ip route 120.0.0.0 255.0.0.0 45.0.0.1  
Router(config)#ip route 23.0.0.0 255.0.0.0 45.0.0.1  
Router(config)#exit  
Router#  
%SYS-5-CONFIG_I: Configured from console by console  
  
Router#|
```

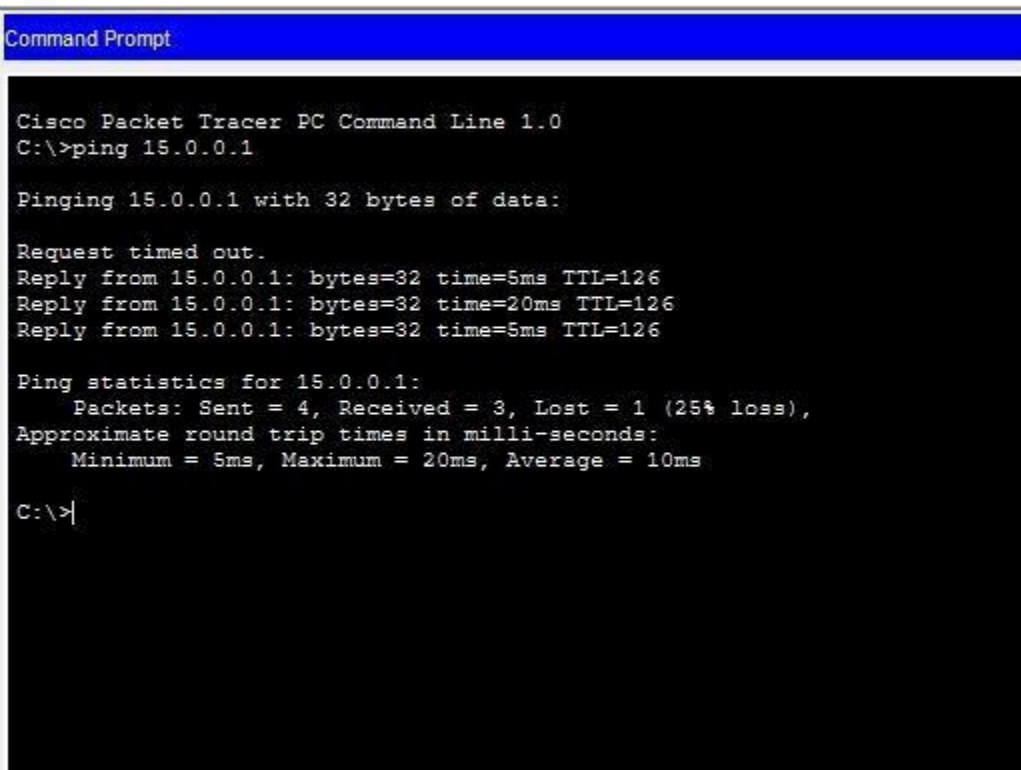
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## Ping:

### PC0 To PC5



## PC1 To PC2

 PC1  


```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 15.0.0.1

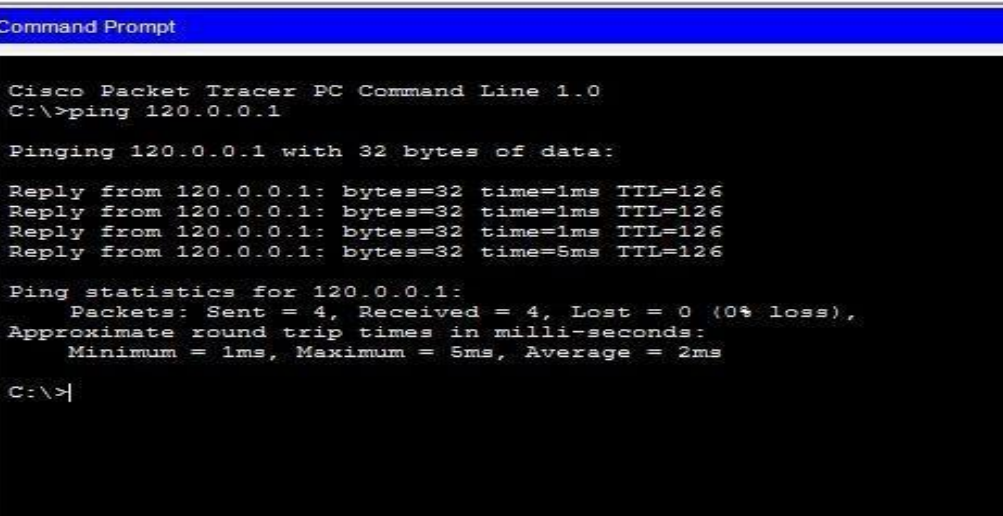
Pinging 15.0.0.1 with 32 bytes of data:

Request timed out.
Reply from 15.0.0.1: bytes=32 time=5ms TTL=126
Reply from 15.0.0.1: bytes=32 time=20ms TTL=126
Reply from 15.0.0.1: bytes=32 time=5ms TTL=126

Ping statistics for 15.0.0.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 20ms, Average = 10ms

C:\>
```

## PC3 To PC1

 PC3  


```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 120.0.0.1

Pinging 120.0.0.1 with 32 bytes of data:

Reply from 120.0.0.1: bytes=32 time=1ms TTL=126
Reply from 120.0.0.1: bytes=32 time=1ms TTL=126
Reply from 120.0.0.1: bytes=32 time=1ms TTL=126
Reply from 120.0.0.1: bytes=32 time=5ms TTL=126

Ping statistics for 120.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 5ms, Average = 2ms

C:\>
```

## **Conclusion**

A three-router network was successfully designed using Class A IP addresses and static routing. Each router was connected to its own LAN and linked to the others through serial connections. After configuring all interfaces and adding static routes, PCs from different networks were able to communicate with each other using ping. This demonstrates that the network design and routing configuration were correct and fully functional.